

Mohammed Arafat Nadaf

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EDUCATION

Northeastern University, Boston, MA

Expected May 2027

Master of Science in Cybersecurity — NSA/DHS Center of Academic Excellence

Coursework: Network Security, Incident Response, Vulnerability Assessment, Security Risk Management & Compliance

Sinhgad Institute of Technology

Jun 2024

Bachelor of Technology, Computer Science — GPA: 9.67/10 | Most Achieving Student Award

PROFESSIONAL EXPERIENCE

PHN Tech Pvt. Ltd.

AI Security & Agentic Systems Engineer Intern

Jun. 2023 – Jun. 2024

- Built and **pressure-tested agentic systems**: exploited Ollama LLM group permission misconfiguration identifying **trust failures** where **authorized** AI agents executed **untrustworthy** actions including model exfiltration and **prompt injection** — directly analogous to Parable's core trust scoring problem; analyzed **behavioral signals** and **identity signals** to detect when seemingly authorized actions should not be trusted
- Performed **adversarial testing** on AI agents and networked systems; exploited CVE-2017-0144, CVE-2021-44228, and CVE-2024-12248; built, tested, failed, and iterated rapidly across **103,163** security events using IBM QRadar **SIEM** and Python/Bash **automation**; reduced triage time **35%** by building and shipping detection pipelines that kept running after handoff
- Designed **security integrations** between **IAM, cloud** (AWS), SIEM, and endpoint tools via **REST API** connections; performed **workflow analysis** to identify where trust decisions failed; demonstrated entrepreneurial **builder** mindset — comfortable with ambiguity, self-directed, iterating until the system was secure

IST Management Services

Security Operations Analyst (Part-Time)

Aug. 2025 – Present

- Built **security integrations** and automated **workflow analysis** using Python and Microsoft 365 APIs; managed **IAM** lifecycle and **identity signals** including access reviews and behavioral anomaly escalation; achieved **100%** audit compliance across **4** consecutive quarterly reviews

RELEVANT PROJECTS

BD Alaris 8015 — Adversarial Testing, Trust Failure Analysis & Iteration [[GitHub](#)]

Medical Device Cybersecurity, Prof. Kevin Fu (former FDA Acting Director) — Spring 2026

- **Pressure-tested** a networked autonomous device end-to-end; built **adversarial** exploits for **6** compound trust failures including authentication bypass, disabled **identity** verification, and unauthorized action execution; built, tested, failed, and iterated until every attack path was documented with CVSS scoring and MITRE ATT&CK validation — shipped a complete analysis independently

SOC Operations Lab — Agentic Workflow Analysis & Cloud Security Integration

Network Security, Northeastern University — Fall 2025

- Built end-to-end **agentic** security pipeline on AWS EC2 processing **75,812** events; integrated **cloud** systems, SIEM, and detection tools via **API**; analyzed **behavioral signals** and **workflow** patterns to surface **trust** anomalies; reduced false positive rate **28%** through iterative rule building and adversarial simulation

Symbiosis University — Backend Engineering, REST APIs & IAM Integration

Full Stack Java Development Internship — Feb. 2024 – Jun. 2024

- Built **backend** systems with **REST API** endpoints and **IAM-enforced identity** controls; integrated **security** into workflow from day one; iterated through test-fail-learn cycles achieving **95%** code coverage; applied ML-adjacent logic for access pattern analysis

TECHNICAL SKILLS

Agentic & AI Security: Agentic systems, trust scoring, behavioral signals, adversarial testing, LLM exploitation

Security Engineering: IBM QRadar SIEM, IAM, identity signals, security integrations, REST APIs, Python, Bash, AWS

Backend & Cloud: Java/Spring Boot, REST APIs, MySQL, AWS (EC2/VPC/IAM), Azure, Linux, GitHub, CI/CD concepts

Builder Mindset: End-to-end shipping, adversarial iteration, pressure-testing, ambiguity tolerance, MITRE ATT&CK, CVSS

Certifications: CompTIA Security+ (July 2026) | 1st Place CTF NEU 2025 (175 pts) | IRJMETs Published Researcher